arXiv:2604.08377

skillclaw

Collective Skill Evolution for Multi-User LLM Agent Ecosystems — enabling skills to continuously improve through cross-user interaction trajectories and agentic, evidence-driven updates.

Agent Skills Remain Static After Deployment

Static Paradigm

- ✗ Skills manually installed, never updated from runtime
- ✗ Failures and solutions rediscovered independently
- ✗ Memory-based methods tied to specific instances
- ✗ No system-level knowledge accumulation

Evolving Paradigm

- ✓ Skills improve continuously from interaction
- ✓ Cross-user signals drive collective evolution
- ✓ Generalizable improvements, not idiosyncratic fixes
- ✓ Knowledge accumulates at the ecosystem level

Closed-Loop Pipeline: Multi-User Interaction → Shared Skill Evolution

U1 U2 U3 U4 U5

▼

① Evidence
Analyze trajectories for behavioral patterns

→

② Attribution
Diagnose root causes of failures & successes

→

③ Evolution
Propose skill mutations via open-ended reasoning

→

④ Distribution
Verify & sync updated skills to all agents

▼

↻ CLOSED-LOOP — CONTINUOUS SKILL SYNCHRONIZATION BACK TO AGENTS

Agentic Evolver: Three-Stage Pipeline

01

Evidence

Analyze session trajectories grouped by referenced skill. Identify behavioral patterns across diverse user contexts.

Preserves full action-feedback causal chains

02

Attribution

Diagnose root causes of observed failures and successes. Separate generalizable signals from idiosyncratic noise.

Context-aware, not rule-based diagnosis

Cross-User Trajectory Aggregation

Different users exercising the same skill under diverse contexts produce complementary views of that skill's behavioral boundary — revealing both conditions under which it works and those under which it breaks.

A single user rarely generates enough signal to separate a **generalizable improvement** from an idiosyncratic fix.

OpenClaw CoPaw IronClaw PicoClaw ZeroClaw NanoClaw NemoClaw

Case Study: Multimodal Creative Pipeline

6-day evolution log — conservative verification prevents regressions

Day	Candidate Skill	Outcome	Rationale
1	validate-tmp-workspace-inputs	Accepted	Check /tmp_workspace inputs & env setup → upgrade to new best pool
2	multimodal-input-validation-and-setup	Rejected	Multimodal input validation & output env init → keep current best pool
3	multimodal-creative-task-pipeline	Rejected	Extract from PDF/video/image pipeline → keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Rejected	Added image classification, visual generation → keep current best pool
5	multimodal-creative-task-pipeline + validate-required-input-files	Rejected	Creative pipeline + per-file fail-fast validation → keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Rejected	Extended PDF-to-poster / document-to-visual → not admitted to next cycle

Case Study: Git Push Auth-Fallback Skill

6-day evolution log — iterative refinement before plateau

Day	Candidate Skill	Outcome	Rationale
1	git-push-with-auth-fallback	Accepted	Safe fallback instead of blocking on push failure → add to Safety best pool
2	git-push-with-auth-fallback	Accepted	Unified patch/bundle filenames, reduced inconsistency → keep updated pool
3	git-push-with-auth-fallback + git-clone-to-directory	Accepted	Auth-alternative paths & secrets audit; fixed subshell pitfalls
4	(retest — same pool)	Accepted	Validator confirmed current pool as best → continue same best pool
5	git-push-with-auth-fallback	Rejected	"Push hang treated as auth failure" — no improvement over current pool
6	git-push-with-auth-fallback	Rejected	Added identity config & filename consistency — no improvement; plateau reached

3 of 6 accepted — successful iterative refinement, then plateau

Three Properties That Distinguish SkillClaw

Collective Evolution

Individual interactions contribute to a shared, continuously improving skill ecosystem. Knowledge accumulates at the system level, not per-user.

⚡ Fully Automatic

Skill evolution is driven entirely by runtime interaction — no manual curation or explicit user intervention required.